

Common tea formulations modulate in vitro digestive recovery of green tea catechins.

Epidemiological evidence suggests a role for tea catechins in reduction of chronic disease risk. However, stability of catechins under digestive conditions is poorly understood. The objective of this study was to characterize the effect of common food additives on digestive recovery of tea catechins. Green tea water extracts were formulated in beverages providing 4.5, 18, 23, and 3.5 mg per 100 mL epicatechin (EC), epigallocatechin (EGC), epigallocatechin-gallate (EGCG), and epicatechin-gallate (ECG), respectively. Common commercial beverage additives; citric acid (CA), BHT, EDTA, ascorbic acid (AA), milk (bovine, soy, and rice), and citrus juice (orange, grapefruit, lemon, and lime) were formulated into finished tea beverages at incremental dosages. Samples were then subjected to in vitro digestion simulating gastric and small intestinal conditions with pre- and post-digestion catechin profiles assessed by HPLC. Catechin stability in green tea was poor with <20% total catechins remaining post-digestion. EGC and EGCG were most sensitive with less, not double equals 10% recovery. Teas formulated with 50% bovine, soy, and rice milk increased total catechin recovery significantly to 52, 55, and 69% respectively. Including 30 mg AA in 250 mL of tea beverage significantly ($p<0.05$) increased catechin recovery of EGC, EGCG, EC, and ECG to 74, 54, 82, and 45% respectively. Juice preparation resulted in the highest recovery of any formulation for EGC (81-98%), EGCG (56-76%), EC (86-95%), and ECG (30-55%). These data provide evidence that tea consumption practices and formulation factors likely impact catechin digestive recovery and may result in diverse physiological profiles.